Student Case Study – Git & GitHub (HTML UI Project)

Scenario
You have joined BrightWeb Solutions as a Graduate Associate. Your first task is to build a simple static website for a company using HTML. Your team will use Git and GitHub to track changes and collaborate on the project.

Project Structure

CompanyWebsite/
■■■ index.html
■■■ about.html
■■■ contact.html
■■■ css/style.css
■■■ images/logo.png (optional)
■■■ README.md

Questions

1. Create the CompanyWebsite project folder and initialize it as a Git repository.

2. Create index.html, about.html, contact.html, css/style.css, and README.md. Check the repository status. What do you observe?

3. Stage only index.html and README.md. How do you verify that only these two files are staged?

4. Stage the remaining files and create your first commit with a meaningful message.

5. Update index.html by adding a company title, navigation menu, and welcome message. Which command shows your changes before committing?

6. Commit the UI changes with an appropriate commit message.

7. Create a new branch named feature-homepage and switch to it.

8. Improve the homepage by adding a footer and an image placeholder. Commit the changes.

9. Switch back to the main branch. Is the new homepage visible? Explain your observation.

10. Merge the feature-homepage branch into the main branch.

11. Create a GitHub repository named CompanyWebsite and connect your local repository.

12. Push your project to GitHub and verify that all HTML files are available online.

13. Create another branch named feature-contact. Update contact.html by adding a contact form (Name, Email, Message). Commit your changes.

14. Push the feature-contact branch and create a Pull Request on GitHub.

15. Merge the Pull Request into the main branch.

16. Display the complete commit history using a graph view. Identify the merge commit.

17. Explain why branches are useful when different developers work on about.html and contact.html simultaneously.

18. Describe one situation that could cause a merge conflict in index.html.

19. List three Git commands you used during this exercise and explain their purpose.

20. Submit your GitHub repository URL and screenshots of git status, branch list, commit history, Pull Request, and the final website.

Submission Checklist

- GitHub Repository URL
- Screenshot of git status
- Screenshot of branch list
- Screenshot of commit history
- Screenshot of Pull Request
- Screenshot of the final HTML website in a browser